

Lecture 2/Module 1

DC Circuit Analysis and Theorems

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Mode of Evaluation:



Mode	Syllabus	Exam Time
CAT I	Module 1,2 (CO 1,2)	August 09-16
Quiz 1	Module 3 (CO 3)	Early Sept
CAT II	Module 3,4 (CO 3,4)	Sept 27- Oct 04
Quiz 2	Module 5 (CO 5)	Mid Oct
Course Project	Module 4 (CO 4)	03 November 2026, Last Instruction Day of Semester
FAT	All Modules	Nov-Dec

CO: Course Outcome (please refer to the Lecture 01)

Last lecture



- Electrical Network and Circuit;
- Basic Circuit Elements and Sources;
- Voltage, Current, Power

Circuit Elements



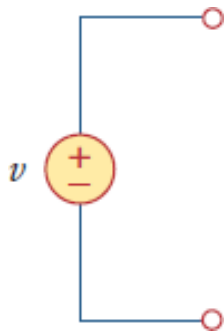
- **Element:** An element is the basic building block of a circuit.
- **Circuit:** An electric circuit is an interconnection of the elements.
- **Circuit Analysis:** It is the process of determining voltages across (currents through) the elements of the circuit.
- **Types of elements:** Active and Passive elements.
- **Active elements:** Capable of generating energy.
Eg.: Generators, batteries, electronic devices and operational amplifiers.
- **Passive elements:** Does not generate energy.
Eg.: Resistors, Capacitors and Inductors.

Independent Sources

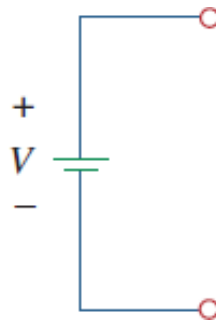


- **Voltage** and **Current** sources are the active elements that generally deliver power to the circuit.
- **Types:** Independent and Dependent sources

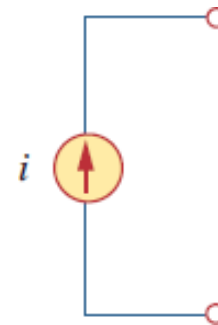
An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.



Constant or Time varying voltage



Constant DC voltage



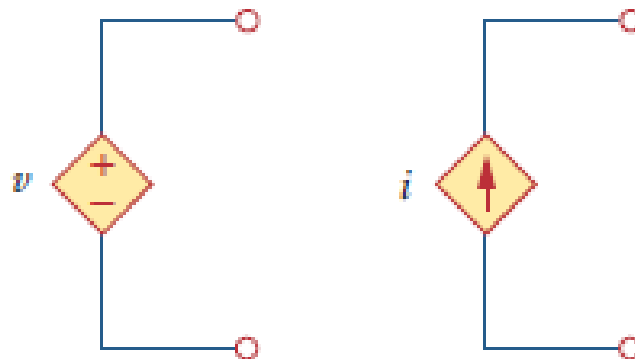
Current source

Dependent Sources



An **ideal dependent** (or **controlled**) **source** is an active element in which the source quantity is controlled by another voltage or current.

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).



Electric Charge, Potential



- The concept of charge
 - The Coulomb [C] – the SI unit of charge
 - An electron carries $e=1.6022 \times 10^{-19}$ C
 - Conservation of charge
- The concept of potential
 - Attraction/repulsion of charges
 - The electric field
 - The energy for moving a charge in a field

Voltage



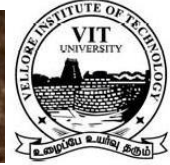
- **Voltage** is a *difference* in electric potential
 - always taken between two points.
 - Absolute voltage is a nonsensical fiction.
 - The concept of *ground* is also a (useful) fiction.
- It is a line integral of the force exerted by an electric field on a unit charge.
- Customarily represented by v or V .
- The SI unit is the Volt [V].

Current



- **Current** is a *movement* of charge.
- It is the time derivative of charge passing through a circuit branch.
- Customarily represented by i or I .
- The SI unit is the Ampere [A].

Power



- **Power** is the product of voltage by current.
- It is the time derivative of energy delivered to or extracted from a circuit branch.
- Customarily represented by P or W .
- The SI unit is the Watt [W].

Charge and Current



Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).

Electric current is the rate of change of charge with respect to time, measured in amperes (A).

$$i = \frac{dq}{dt}$$

1 ampere = 1 coulomb/second

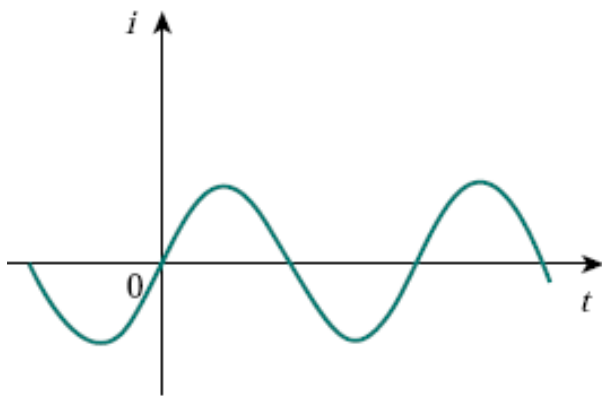
Charge and Current



(a)

If the current does not change with time, but remains constant, we call it a *direct current (dc)*.

A direct current (dc) is a current that remains constant with time.



(b)

A time-varying current is represented by the symbol i . A common form of time-varying current is the sinusoidal current or *alternating current (ac)*.

An alternating current (ac) is a current that varies sinusoidally with time.

Voltage



Voltage is the energy per unit charge.

$$v = \frac{dw}{dq} \quad \dots \text{definition of magnitude}$$

v = the voltage in volts

w = the energy in joules

q = the charge in coulombs

($e = 1.6022 \times 10^{-19}$ C)

Power



Power is the energy per unit time.

$$p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = v \times i.$$

p = the power in watts

w = the energy in joules

t = the time in seconds

v = voltage in volts

i = current in amperes

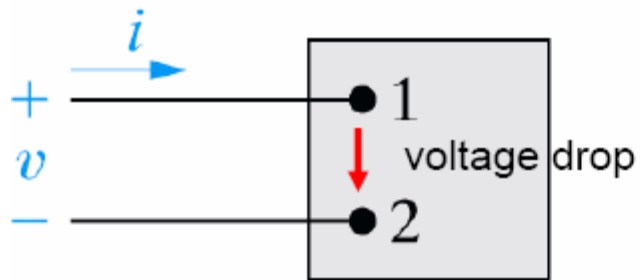
Power dissipation and extraction



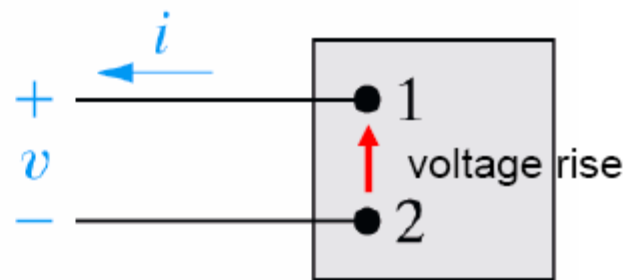
As positive charges experience voltage drop, they lose energy.

- **$p > 0$ (not $p = +vi$): power is dissipated by the circuit inside the box.**
- **$p < 0$ (not $p = -vi$): power is being extracted.**

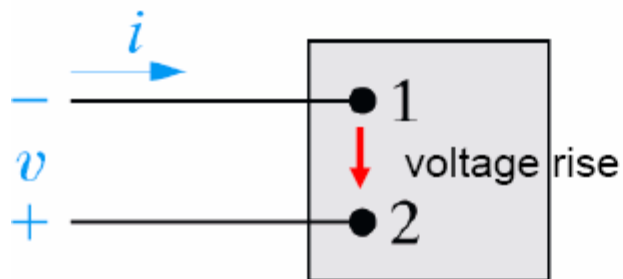
Polarity references and power expressions



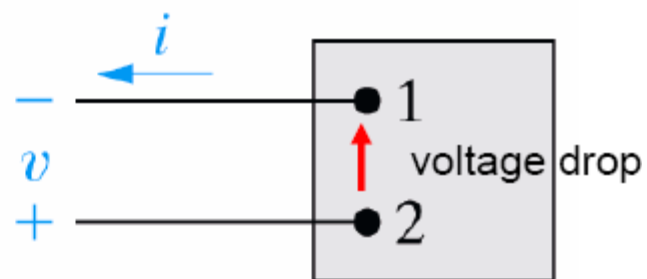
(a) $p = vi$



(b) $p = -vi$



(c) $p = -vi$



(d) $p = vi$

Lecture Topics



- Ohm's Law; Node, Branch, Loops

Ohm's Law



Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

$$v \propto i$$

$$v = iR$$

The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

$$R = \frac{v}{i}$$

Resistance

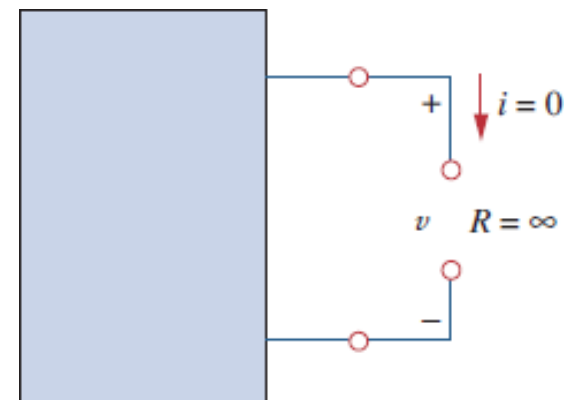
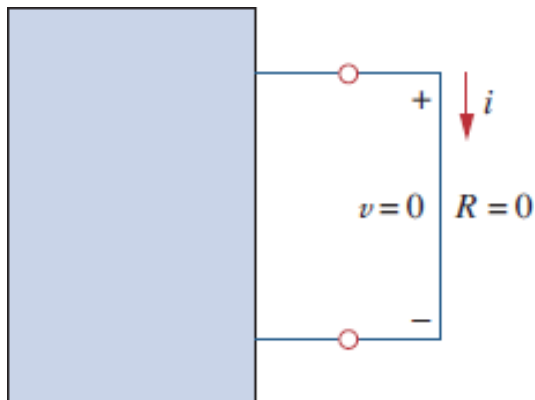


A **short circuit** is a circuit element with resistance approaching zero.

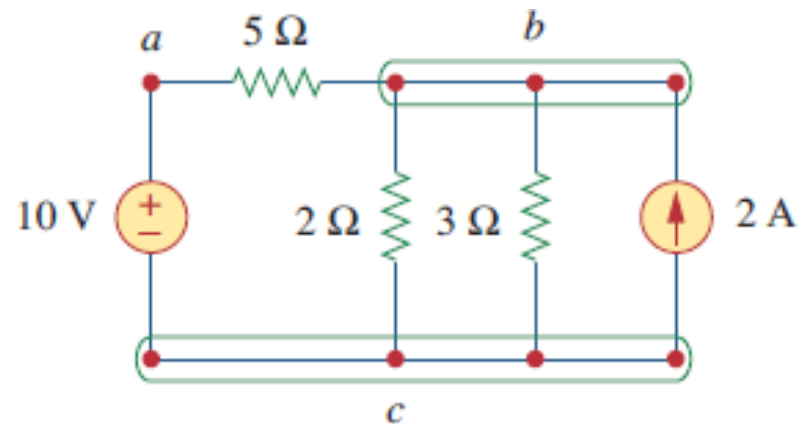
$$v = iR = 0$$

An **open circuit** is a circuit element with resistance approaching infinity.

$$i = \lim_{R \rightarrow \infty} \frac{v}{R} = 0$$



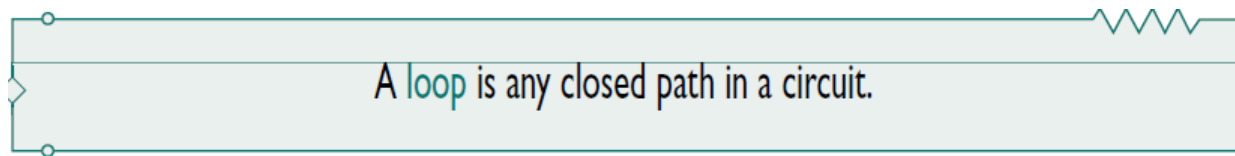
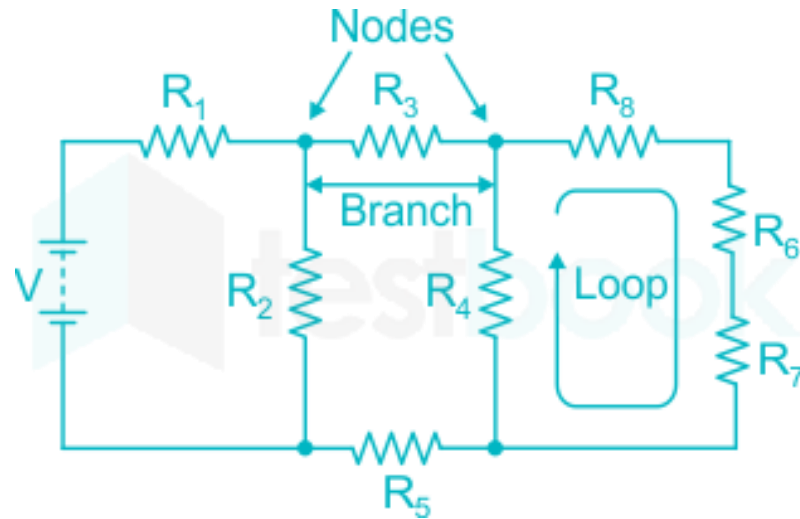
Circuit



A **branch** represents a single element such as a voltage source or a resistor.

A **node** is the point of connection between two or more branches.

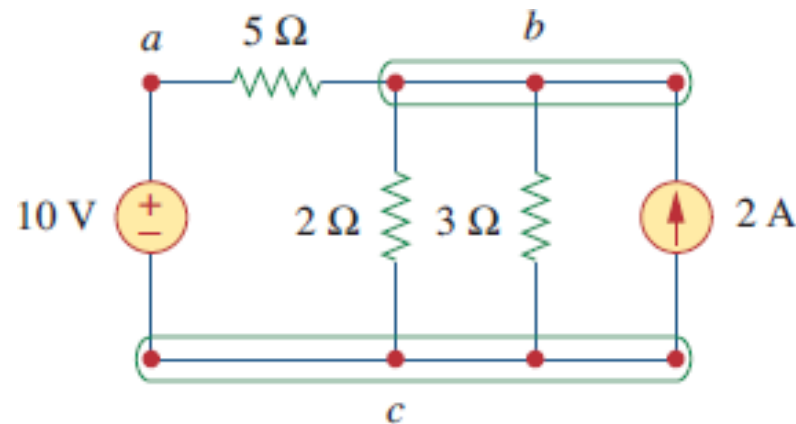
Circuit



An **independent Loop** is a loop that doesn't contain in any other loop inside

Independent Loop is also called **Mesh**

Circuit

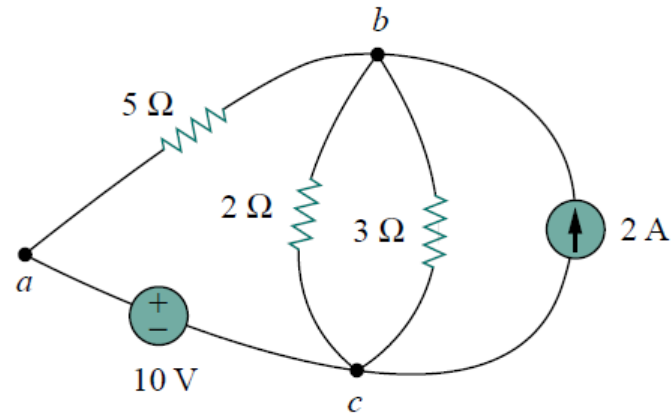


Let's consider the above circuit, if b , n and l represent number of Branches, Nodes and Independent Meshes respectively, then

$$b = l + n - 1$$

Can you realize this relationship for the above circuit?? Find b , n and l .

Circuit



Do the same exercise for the above circuit. Find b , n and l and check the relationship between b , n and l holds true or not for this circuit.

Thank You